

The Byzantine Generals Problem

Understanding Byzantine Faults and $OM(m)$

Your Name

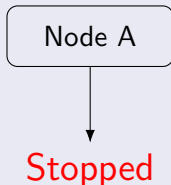
July 11, 2026

Presentation Overview

- Motivation: why Byzantine faults matter
- The Byzantine Generals Problem
- Interactive consistency
- Why three generals are not enough
- $OM(m)$: Oral Messages algorithm
- Connection to PBFT

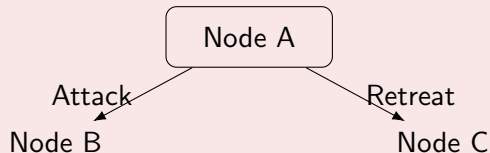
Crash Fault vs Byzantine Fault

Crash Fault



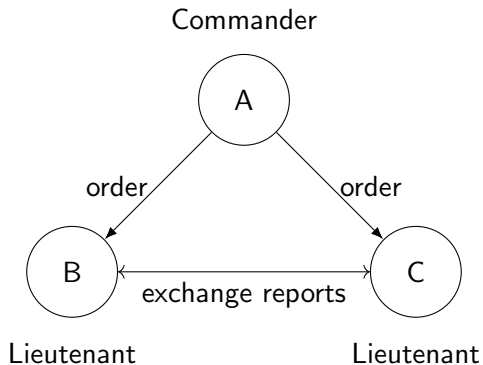
- Node stops responding
- Easier to detect
- Paxos / Raft mainly assume this model

Byzantine Fault



- Node may lie
- Sends inconsistent messages
- Much harder to tolerate

The Byzantine Generals Problem



Goal

All loyal generals must choose the same action: **Attack** or **Retreat**.

IC1: Agreement

All loyal lieutenants obey the same order.

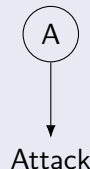
B: Attack

C: Attack

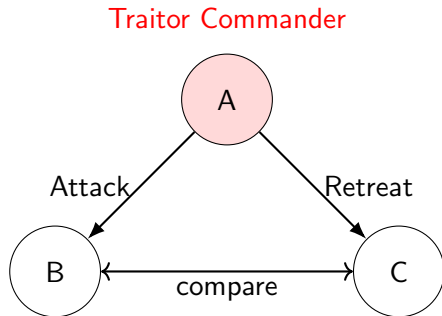
D: Attack

IC2: Validity

If the commander is loyal, every loyal lieutenant follows the commander's order.



Why Three Generals Are Impossible



Problem

B sees "Attack" from A and "Retreat" from C. B cannot distinguish whether A is lying or C is lying.

Information Ambiguity

World	B's Observation	Possible Explanation
World 1	A says Attack, C says Retreat	A is traitor
World 2	A says Attack, C says Retreat	C is traitor

Key Insight

The loyal general cannot distinguish two different worlds from the same local information.

Conclusion

With three generals and one traitor, no algorithm can guarantee both IC1 and IC2.

How Many Generals Are Needed?

Required Number

To tolerate m Byzantine traitors:

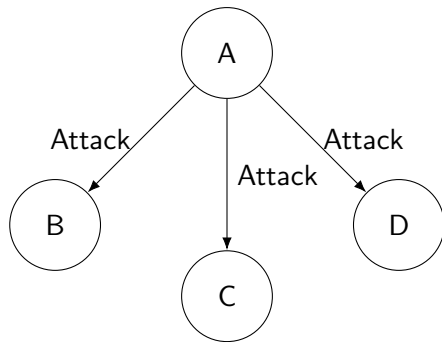
$$n \geq 3m + 1$$

Traitors m	Required Generals
0	1
1	4
2	7
3	10

Connection

PBFT uses the same principle: to tolerate f Byzantine replicas, it requires $3f + 1$ replicas.

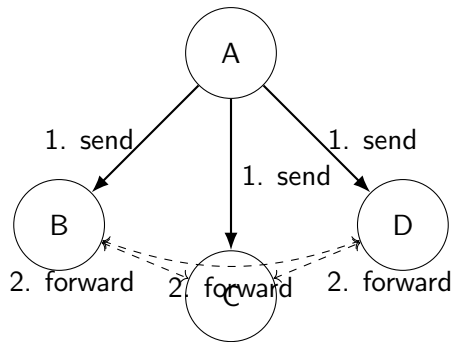
OM(0): No Traitors



OM(0)

The commander sends a value, and each lieutenant adopts the received value.

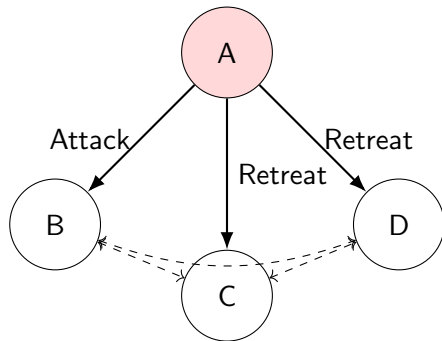
OM(1): One Traitor



OM(1)

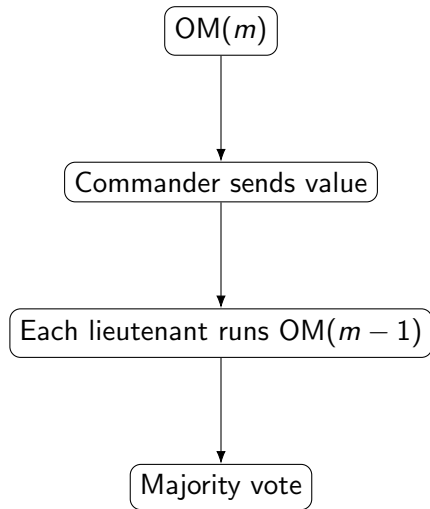
Each lieutenant forwards what it received, then takes a majority vote.

OM(1) Example: Commander Is Traitor



Lieutenant	Direct	From B	From C	Decision
B	Attack	–	Retreat	Retreat
C	Retreat	Attack	–	Retreat
D	Retreat	Attack	Retreat	Retreat

General Form of $OM(m)$



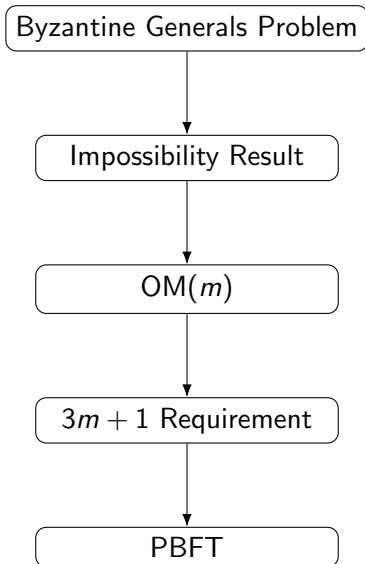
Message Depth in $OM(m)$

Algorithm	Faults Tolerated	Message Depth
$OM(0)$	0	1 step
$OM(1)$	1	2 steps
$OM(2)$	2	3 steps
$OM(m)$	m	$m + 1$ steps

Intuition

More Byzantine faults require deeper message forwarding, because loyal nodes need more evidence from multiple paths.

From Byzantine Generals to PBFT



PBFT Interprets the Theory Practically

Byzantine Generals

- Theoretical model
- Oral messages
- Recursive forwarding
- Agreement among loyal generals

PBFT

- Practical replicated service
- Pre-prepare
- Prepare
- Commit
- View change

Key Difference

PBFT is not just majority voting. It orders client requests safely in a real distributed system.

Takeaways

- Byzantine faults are stronger than crash faults.
- The goal is agreement among loyal nodes, not necessarily identifying traitors.
- With three generals and one traitor, agreement is impossible.
- OM(m) uses recursive message forwarding and majority voting.
- To tolerate m Byzantine faults, at least $3m + 1$ generals are required.
- PBFT builds on this theory to implement practical Byzantine fault-tolerant replication.